

Lecture 2 — Analysis of Randomized Algorithms; Sorting & Searching

Course **2-INF-135/15 Pravdepodobnostné algoritmy**, LS 2025/26. Source slides: [RA_02.pdf](#) (16 pages).

What this lecture is about

Lecture 1 was a *gallery* — many algorithms, each analyzed with whatever trick fit. This lecture stops to build the **standard toolbox** that every later analysis reuses, and then spends it on two case studies.

It has two halves:

1. **The probability toolkit** — three *tail inequalities* (Markov, Chebyshev, Chernoff) that answer the recurring question "*how unlikely is it that a random quantity strays far from its average?*" This is the property called **concentration** (koncentrácia).
2. **Sorting & searching** — two payoffs:
 - **Randomized QuickSort runs in $O(n \log n)$ not just on average but with high probability** — a much stronger guarantee than Lecture 1's expectation bound.
 - **Randomized Selection (R-Select)** finds the median (or any rank) in $2n + o(n)$ comparisons — *linear* time, beating the $n \log n$ of sorting.

The spine of the whole lecture is one idea: **the three inequalities form a ladder. Each rung assumes more about the random variable and pays you back with a sharper bound.** Get that ladder straight and everything else follows.

1. The probability vocabulary (recap, with the right names)

A quick refresher of the objects we keep using. Let X be a **random variable** (náhodná premenná), and let E_i be the event " $X = i$ ".

- **Expected value** (stredná hodnota), also written μ : $E[X] = \sum_i i \cdot \Pr[X = i]$.
- **Variance** (rozptyl) and **standard deviation** (odchýlka) σ :
$$\text{Var}(X) = \sum_i (i - E[X])^2 \Pr[X = i] = E[(X - E[X])^2], \quad \sigma = \sqrt{\text{Var}(X)}.$$
 Variance measures *how spread out* X is around its mean.

- **Union bound** (the most-used inequality in the course): $\Pr \left[\bigcup_i E_i \right] \leq \sum_i \Pr[E_i]$. When is it an equality? Exactly when the events are **pairwise disjoint** (no overlap is double-counted).
- **Linearity of expectation** — the engine of Lecture 1, holds **with or without independence**:

$$E \left[\sum_i a_i X_i \right] = \sum_i a_i E[X_i].$$

Independence and what it unlocks

X, Y are **independent** (nezávislé) if

$$\Pr[X = x, Y = y] = \Pr[X = x] \Pr[Y = y] \iff \Pr[X = x \mid Y = y] = \Pr[X = x].$$

(k -independence is the same statement for every k -tuple — this is the notion Lecture 5's

derandomization will weaken on purpose.)

Independence buys two things linearity does **not**:

$$E[XY] = E[X] E[Y], \quad \text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y].$$

Watch the asymmetry. Expectation *adds* for free (linearity, always). Variance *adds* only under independence. That single fact is why the whole Chebyshev machinery below needs independence and the linearity arguments of Lecture 1 did not.

Two estimates we will reach for constantly

$$1 + x \leq e^x \quad (\forall x), \quad \left(1 - \frac{1}{n}\right)^n \leq \frac{1}{e} \leq \left(1 - \frac{1}{n}\right)^{n-1}.$$

The first ($1 + x \leq e^x$, equivalently $1 + y < e^y$) is the workhorse — it is exactly the step that turns a product into a clean exponential in the Chernoff proof.

"With high probability" (s vysokou pravdepodobnosťou, s.v.p.)

An event happens **with high probability** if its probability is $1 - O\left(\frac{1}{n^c}\right)$ for some constant $c > 0$.

This is the target we aim for. The exponent c matters: if we later want to **union-bound over n bad events** (e.g. n root-to-leaf paths in a sort tree), each must fail with probability $\leq 1/n^2$ so the *total* failure is $\leq n \cdot 1/n^2 = 1/n$ — still high probability. Keep that bookkeeping in mind; it dictates *how* sharp a tail bound we need.

2. The running example — balls into boxes (guličky do krabíc)

To compare the three inequalities we keep one concrete question in front of us:

Throw m balls independently and uniformly into n boxes. Let X = the number of balls in **one fixed box**. Then $E[X] = m/n$. **How likely is that box to be much fuller than average?**

X is **binomial**: $X \sim \text{Bin}(m, 1/n)$, so

$$E[X] = \frac{m}{n} = \mu, \quad \text{Var}(X) = m \cdot \frac{1}{n} \left(1 - \frac{1}{n}\right) = \mu \left(1 - \frac{1}{n}\right) < \mu.$$

We will ask the *same* question — *what is $\Pr[X \geq 2\mu]$?* — of all three inequalities and watch the answer get sharper each time.

3. Markov's inequality — the root of everything

Theorem (Markov). If $X \geq 0$ (takes only non-negative values), then for every $k > 0$
 $\Pr[X \geq k] \leq \frac{E[X]}{k}$, equivalently $\Pr[X \geq k E[X]] \leq \frac{1}{k}$.

Proof — one line of bookkeeping. Throw away every term below k and underestimate the rest by k : $E[X] = \sum_i i \Pr[X = i] \geq \sum_{i \geq k} k \Pr[X = i] = k \sum_{i \geq k} \Pr[X = i] = k \Pr[X \geq k]$. Divide by k . ■

That's the whole thing. Markov knows **only the mean** and **only that X can't go negative**. With so little information it can't say much — but it is the seed from which the other two grow.

On the balls: $\Pr[X \geq 2\mu] \leq \frac{\mu}{2\mu} = \frac{1}{2}$. Almost useless, but honest.

The deep point to say aloud. Markov is *the* tail inequality. Chebyshev and Chernoff are not new ideas — they are **Markov applied to a cleverly transformed version of X** . Keep that in mind through the next two sections.

4. Chebyshev's inequality — Markov on the squared deviation

Markov is weak because it ignores the *shape* of X . Chebyshev feeds it one more number — the **variance** — and in return gets a **two-sided** bound that needs **no non-negativity assumption**.

Theorem (Chebyshev). For any random variable X and every $k > 0$,
 $\Pr [|X - E[X]| \geq k] \leq \frac{\text{Var}(X)}{k^2}$, equivalently $\Pr [|X - E[X]| \geq k\sigma] \leq \frac{1}{k^2}$.

The second form is the memorable one: *the probability of being k standard deviations off the mean is at most $1/k^2$.*

Proof — it really is just Markov. The deviation can be negative, so square it to make it non-negative, then apply Markov to $Y = (X - E[X])^2$:

$\Pr [|X - E[X]| \geq k] = \Pr [(X - E[X])^2 \geq k^2] \leq \frac{E[Y]}{k^2} = \frac{\text{Var}(X)}{k^2}$. The last equality is just the definition $E[(X - E[X])^2] = \text{Var}(X)$. ■

On the balls. Use $\text{Var}(X) = \mu(1 - 1/n) < \mu$:

$\Pr[X \geq 2\mu] \leq \Pr[|X - \mu| \geq \mu] \leq \frac{\text{Var}(X)}{\mu^2} < \frac{\mu}{\mu^2} = \frac{1}{\mu} = \frac{n}{m}$.

Sharper already. Markov gave $\frac{1}{2}$ no matter what; Chebyshev gives n/m , which is *tiny* when there are many balls per box ($m \gg n$). The price was one extra assumption — that we know (and can bound) the variance.

5. Chernoff bound — Markov on the exponential

For a **sum of independent** indicators we can do dramatically better. Knowing the tail of such a sum exactly is hopeless — the exact formula is a sum over all large subsets,

$\Pr[X \geq k] = \sum_{\substack{A \subseteq \{1, \dots, n\} \\ |A| \geq k}} \prod_{i \in A} p_i \prod_{i \notin A} (1 - p_i)$, which has exponentially many terms. Chernoff

replaces that monster with a clean **exponentially small** bound.

Theorem (Chernoff). Let $X_1, \dots, X_n$ be **independent** 0/1 variables with $p_i = \Pr[X_i = 1]$, let $X = \sum_i X_i$ and $\mu = E[X] = \sum_i p_i$. Then:

$\forall \delta > 0 : \Pr[X \geq (1 + \delta)\mu] < \left(\frac{e^\delta}{(1 + \delta)^{1 + \delta}}\right)^\mu \quad (\star)$

$\forall \delta \in (0, 1) : \Pr[X \geq (1 + \delta)\mu] \leq e^{-\mu\delta^2/3} \quad \forall \delta \in (0, 1) : \Pr[X \leq (1 - \delta)\mu] \leq e^{-\mu\delta^2/2}$

$\forall R \geq 6\mu : \Pr[X \geq R] \leq 2^{-R}$

The middle two are the **usable everyday forms** — "the probability of being a δ -fraction off the mean decays like $e^{-\mu\delta^2}$." The key feature: the bound shrinks **exponentially in μ** , not polynomially.

The proof — the "MGF + Markov" trick (worth memorizing)

The recipe is three moves: **make it non-negative** → **exponentiate** → **Markov**.

Step 1 — exponentiate, then Markov. e^{tX} is non-negative for any $t > 0$, and $x \mapsto e^{tx}$ is increasing, so the event $X \geq (1 + \delta)\mu$ is the same event as $e^{tX} \geq e^{t(1+\delta)\mu}$. Apply Markov to e^{tX} : $\Pr[X \geq (1 + \delta)\mu] = \Pr[e^{tX} \geq e^{t(1+\delta)\mu}] \leq \frac{E[e^{tX}]}{e^{t(1+\delta)\mu}}$.

Step 2 — the moment generating function factorizes (here is where **independence** is spent):

$E[e^{tX}] = E[e^{t\sum_i X_i}] = E\left[\prod_i e^{tX_i}\right] \stackrel{\text{indep.}}{=} \prod_i E[e^{tX_i}] = \prod_i (p_i e^t + (1 - p_i))$. Now use $1 + y < e^y$ with $y = p_i(e^t - 1)$: $\prod_i (1 + p_i(e^t - 1)) < \prod_i e^{p_i(e^t - 1)} = e^{(e^t - 1)\sum_i p_i} = e^{\mu(e^t - 1)}$.

Step 3 — optimize t . Substituting back, $\Pr[X \geq (1 + \delta)\mu] < \frac{e^{\mu(e^t - 1)}}{e^{t(1+\delta)\mu}} = e^{(\mu(e^t - 1) - t(1+\delta)\mu)}$. Minimize the exponent over t : the derivative gives $e^t = 1 + \delta$, i.e. $t = \ln(1 + \delta)$. Plugging in yields exactly (★): $\Pr[X \geq (1 + \delta)\mu] < \left(\frac{e^\delta}{(1+\delta)^{1+\delta}}\right)^\mu$. ■ The friendlier forms ($e^{-\mu\delta^2/3}$ etc.) come from bounding this expression for δ in the stated ranges.

On the balls. With $\mu = m/n$ and $\delta = 1$: $\Pr[X \geq 2\mu] \leq e^{-\mu\delta^2/3} = e^{-m/(3n)}$.

The escalation, side by side (same event, $\Pr[X \geq 2\mu]$ for the fullest box):

$\underbrace{\frac{1}{2}}_{\text{Markov}} \gg \underbrace{\frac{n}{m}}_{\text{Chebyshev}} \gg \underbrace{e^{-m/(3n)}}_{\text{Chernoff}}$. Constant → polynomially small → **exponentially** small.

Each rung cost one more assumption.

6. The ladder (the unifying idea — say this in the oral)

The three inequalities are **one idea at three resolutions**. Read the table top to bottom: each row assumes strictly more and pays back a strictly sharper tail.

Inequality	Needs	Mechanism	Tail decay
Markov	$X \geq 0$, the mean	— (direct)	$\sim 1/k$
Chebyshev	the variance	Markov on $(X - \mu)^2$	$\sim 1/k^2$
Chernoff	independence of a sum	Markov on e^{tX}	$\sim e^{-k}$

The one sentence that ties it together. Chebyshev and Chernoff are both Markov in disguise — applied not to X but to a transformed variable that amplifies the tail before Markov sees it. Squaring $((X - \mu)^2)$ turns a two-sided question into a one-sided non-negative one and earns a $1/k^2$. Exponentiating (e^{tX}) is even more aggressive: it blows the tail up so violently that, after optimizing the knob t , what survives decays **exponentially**. Stronger transform $\rightarrow$ stronger bound — but e^{tX} only factorizes when the terms are **independent**, which is the price Chernoff pays and the other two don't.

7. RQS is $O(n \log n)$ with high probability

Lecture 1 proved randomized QuickSort uses $\approx 2n \ln n$ comparisons **in expectation**. That is an average — a single run could (rarely) be much worse. Now we upgrade to a **with-high-probability** guarantee: *almost every* run is $O(n \log n)$, not just the average run. Chernoff is exactly the tool for this.

Good pivots vs. bad pivots

Model a run as the **recursion tree** $\text{RQS}(S)$: root S , children $S_<$ and $S_>$ (the elements below / above the pivot). Total work is $O(n \cdot \text{depth})$, so **it suffices to bound the depth**, i.e. the length of the longest root-to-leaf path.

Call a pivot **good** if it splits its set in a roughly balanced way: $|S_<|, |S_>| \leq \frac{2}{3}|S|$.

A pivot is good iff it lands in the middle third of the sorted order, so

$$\Pr[\text{pivot is good}] = \frac{1}{3}, \quad \Pr[\text{bad}] = \frac{2}{3}.$$

Why good pivots cap the depth. Each good pivot shrinks the set by a factor $\leq \frac{2}{3}$. If $S, S_1, S_2, \dots$ are the sets where good pivots occurred along a path, then $|S_i| \leq \left(\frac{2}{3}\right)^i |S|$. The set hits size 1 after at most $c \log n$ good pivots, $c = \frac{1}{\ln(3/2)} \approx 2.43$ (good pivots on a path $\leq c \ln n$). So a path can only be *long* if it is stuffed with **bad** pivots — and bad pivots are where the randomness can be pinned down by Chernoff.

The Chernoff step

Fix a single root-to-leaf path and look at its first $60 \ln n$ vertices. Let

$$X_i = \begin{cases} 1 & \text{\textit{i-th vertex on the path has a bad pivot}} \\ 0 & \text{otherwise} \end{cases}, \quad \Pr[X_i = 1] = \frac{2}{3}, \text{ and}$$

$$X = \sum_{i=1}^{60 \ln n} X_i, \text{ so } \mu = E[X] = \frac{2}{3} \cdot 60 \ln n = 40 \ln n.$$

A path with **more than $60 \ln n$** vertices must contain at least $60 \ln n - 2.43 \ln n = 57.57 \ln n$ bad ones (only $\leq 2.43 \ln n$ can be good). So the path being too long forces X far above its mean:
 $\Pr[\text{path} > 60 \ln n] \leq \Pr[X \geq 57.57 \ln n] < \Pr[X \geq 56 \ln n] = \Pr\left[X \geq \left(1 + \frac{2}{5}\right) \underbrace{40 \ln n}_{\mu}\right]$. With $\delta = \frac{2}{5}$, the Chernoff form $\Pr[X \geq (1 + \delta)\mu] \leq e^{-\mu\delta^2/3}$ gives
 $\leq \exp\left(-40 \ln n \cdot \frac{(2/5)^2}{3}\right) = n^{-160/75} < n^{-2}$.

From one path to the whole tree (union bound)

One path is short except with probability $< n^{-2}$. There are at most n leaves, hence $\leq n$ paths. Union bound: $\Pr[\text{some path} > 60 \ln n] \leq n \cdot n^{-2} = \frac{1}{n}$.

So **every** path has length $O(\log n)$ with probability $\geq 1 - 1/n$, i.e. the tree has depth $O(\log n)$ w.h.p., i.e. RQS does $O(n \log n)$ work w.h.p.

Punchline. This is *strictly stronger* than the expectation bound. The argument is the canonical w.h.p. recipe: **(1)** isolate "good" events with a constant success probability; **(2)** Chernoff a single object to make its failure $\leq n^{-2}$; **(3)** union-bound over the n objects to get total failure $\leq 1/n$. The exponent-2 in step (2) is *engineered* precisely so that step (3) survives — this is why we needed Chernoff and not Chebyshev.

8. R-Select — finding the median in $2n + o(n)$ comparisons

Sorting finds the median in $O(n \log n)$. Can we do **linear**? Yes — and the idea is pure sampling: **look at a sublinear random sample, use it to trap the median inside a tiny window, then sort only that window.**

The algorithm (median = the $\frac{n}{2}$ -th element)

Input: set S , $|S| = n$. Output: the median m .

1. **Sample.** Draw $R \leftarrow n^{3/4}$ elements from S , uniformly **with replacement**.
2. **Bracket.** Sort R . Let $\ell = \frac{n^{3/4}}{2} - \sqrt{n}$, $u = \frac{n^{3/4}}{2} + \sqrt{n}$, and let d, h be the ℓ -th and u -th smallest elements of R . These two sampled values are our guessed lower/upper fence around the median.
3. **Filter S against the fences:**
 $D = \{x \in S : x < d\}$, $H = \{x \in S : x > h\}$, $C = \{x \in S : d \leq x \leq h\}$.
4. **Decide.** If $|D| > \frac{n}{2}$ or $|H| > \frac{n}{2}$ or $|C| > 4n^{3/4}$, then **FAIL**. Otherwise the median lies in the small set C — find it there (by sorting C).

Why $2n + o(n)$ comparisons. Sorting R costs $O(n^{3/4} \log n) = o(n)$. Building D, H, C compares each of the n elements against d and h — that's the $2n$. Sorting the surviving window C (size $\leq 4n^{3/4}$) costs $o(n)$ again. The two fence-tests dominate: $2n + o(n)$, genuinely linear.

The three FAIL conditions are exactly the three ways the plan can go wrong, and each is killed by **Chebyshev** (we don't even need Chernoff here — we only need the failure probability to vanish).

Bounding $\Pr[|D| > n/2]$ — Chebyshev on the sample count

$|D| > \frac{n}{2}$ means more than half of S lies below the fence d , i.e. the true median fell *below* d (we bracketed too high). Let $X_i = \begin{cases} 1 & \text{i-th sampled element} \leq m \\ 0 & \text{otherwise} \end{cases}$, $X = \sum_{i=1}^{n^{3/4}} X_i$. Since m is the median, each sample lands $\leq m$ with probability exactly $\frac{1}{2}$, so $X \sim \text{Bin}(n^{3/4}, \frac{1}{2})$ with $E[X] = \frac{n^{3/4}}{2}$, $\text{Var}(X) = \frac{n^{3/4}}{4}$. The bad event " d ended up above m " happens iff fewer than ℓ samples were $\leq m$, i.e. $X < \frac{n^{3/4}}{2} - \sqrt{n}$. Apply Chebyshev with $k = \sqrt{n}$:
 $\Pr[X < \frac{n^{3/4}}{2} - \sqrt{n}] \leq \Pr[|X - E[X]| > \sqrt{n}] \leq \frac{\text{Var}(X)}{(\sqrt{n})^2} = \frac{n^{3/4}/4}{n} = \frac{n^{-1/4}}{4}$. By symmetry
 $\Pr[|H| > \frac{n}{2}] \leq \frac{n^{-1/4}}{4}$ as well.

Bounding $\Pr[|C| > 4n^{3/4}]$ — the window stays small

If the window C is too big, then it bulges on one side of the median:

$|C| > 4n^{3/4} \implies |C_{\leq m}| > 2n^{3/4}$ or $|C_{\geq m}| > 2n^{3/4}$. Take the upper bulge $|C_{\geq m}| > 2n^{3/4}$: it means the fence h sits a full $2n^{3/4}$ ranks above the median, i.e. h is among the top $\frac{n}{2} - 2n^{3/4}$ elements of S . Let $Y_i = \begin{cases} 1 & \text{i-th sample is among the top } \frac{n}{2} - 2n^{3/4} \text{ of } S \\ 0 & \text{otherwise} \end{cases}$, $Y = \sum_{i=1}^{n^{3/4}} Y_i$, so $Y \sim \text{Bin}(n^{3/4}, p)$ with $p = \frac{1}{2} - \frac{2}{n^{1/4}}$:
 $E[Y] = n^{3/4}p = \frac{n^{3/4}}{2} - 2\sqrt{n}$, $\text{Var}(Y) = n^{3/4}p(1-p) < \frac{n^{3/4}}{4}$. The bulge forces $Y > \frac{n^{3/4}}{2} - \sqrt{n}$, which is $\sqrt{n}$ above $E[Y]$. Chebyshev again:
 $\Pr[Y > \frac{n^{3/4}}{2} - \sqrt{n}] = \Pr[Y - E[Y] > \sqrt{n}] \leq \Pr[|Y - E[Y]| > \sqrt{n}] < \frac{n^{-1/4}}{4}$. Two sides, so
 $\Pr[|C| > 4n^{3/4}] < 2 \cdot \frac{n^{-1/4}}{4} = \frac{n^{-1/4}}{2}$.

Putting the three together

Theorem. R-Select finds the median with probability $\geq 1 - n^{-1/4}$.

$$\Pr[\text{FAIL}] \leq \underbrace{\Pr[|D| > \frac{n}{2}]}_{\leq n^{-1/4}/4} + \underbrace{\Pr[|H| > \frac{n}{2}]}_{\leq n^{-1/4}/4} + \underbrace{\Pr[|C| > 4n^{3/4}]}_{\leq n^{-1/4}/2} = n^{-1/4}.$$

On FAIL, just **restart**. Failures are independent, so $\Pr[\text{FAIL after } \ell \text{ runs}] \leq n^{-\ell/4}$, and the expected number of runs is $E[\ell] < 2$. So the expected total cost stays $2n + o(n)$.

Punchline. A *sublinear* sample of size $n^{3/4}$ is enough to pin the median's rank to within $\pm\sqrt{n}$ (the standard-deviation scale of a binomial), which traps the answer in a window of size $O(n^{3/4})$ that we can afford to sort outright. Chebyshev — the *middle* rung of the ladder — is already strong enough, because we only need failure $\rightarrow 0$, not n^{-c} . This $2n + o(n)$ is essentially the optimal comparison constant for selection.

Generalization — the k -th smallest element

The same template returns the k -th smallest S_k for any rank k (not just the median). Only the centering changes: set $x = k/n^{1/4}$ and put the sample fences at $\ell = \max\{\lfloor x - \sqrt{n} \rfloor, 1\}$ and $u = \min\{\lceil x + \sqrt{n} \rceil, n^{3/4}\}$. Near the ends ($k < n^{1/4}$ or $k > n - n^{1/4}$) you keep a one-sided window $\{x \leq h\}$ resp. $\{x \geq d\}$; in the bulk you keep $\{d \leq x \leq h\}$. FAIL if the window grows past the $O(n^{3/4})$ budget. Same guarantee: $\Pr[\text{FAIL}] \leq n^{-1/4}$, cost $2n + o(n)$.

Recurring themes to carry forward

Theme	Where it appeared	Reused later in
The tail-bound ladder (Markov $\subset$ Chebyshev $\subset$ Chernoff)	§3–6	every concentration argument in the course
"Markov on a transformed variable" (square $\rightarrow$ Chebyshev, exponentiate $\rightarrow$ Chernoff)	§4, §5	the meta-trick to remember
MGF + optimize t (the Chernoff proof)	§5	error amplification, expander walks
Chernoff + union bound $\rightarrow$ w.h.p. (engineer n^{-2} per object, sum over n)	§7 RQS depth	any "every one of n things behaves" claim
Variance adds only under independence	§1	why Chebyshev needs independent summands
Sample to estimate ranks (sublinear sample fixes the answer to $\pm\sqrt{n}$)	§8 R-Select	sampling-based algorithms generally
Restart-on-FAIL ($E[\ell] < 2$)	§8	turning a Monte-Carlo failure into Las-Vegas expected cost

The single sentence that ties the lecture together:

Concentration is the whole game: a random quantity almost never strays far from its mean, and the three inequalities are one idea — apply Markov after a transform — at escalating strength. Spend a sharper bound where you must (Chernoff to union-bound RQS over n paths) and a cheaper one where you can (Chebyshev to keep R-Select's sample window small).